

EFFECTS OF LOSARTAN – GLIBENCLAMIDE COMBINATION THERAPY ON THE METABOLISM IN HYPERTENSIVE NIDDM PATIENTS

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ABSTRACT

In order to observe the effects of Losartan glibenclamide combination therapy on the metabolism (-CHO and fats) and physiological factors (Blood pressure, weight and BMI) in hypertensive NIDDM patients, combined drug therapy with tablet losartan 50 mg and tablet glibenclamide 5 mg (adjusted according to the glycemic control of the individual) was given to 10 newly diagnosed uncomplicated hypertensive NIDDM patients of both sexes between 30 to 55 years of age for a period of 5 weeks. Fasting blood sugar and blood pressure levels were taken once weekly whereas fasting serum insulin level, lipid profile (total cholesterol, triglycerides, HDL-cholesterol, LDL-cholesterol, VLDL-cholesterol), weight and body mass index were done twice during the study i.e. at day zero and day 35.

Significant reduction in fasting blood glucose level ($P<0.001$). Significant increase in the fasting serum insulin level ($P<0.05$). Non-significant changes ($P=NS$) in the total serum cholesterol, triglyceride, LDL and VLDL cholesterol levels. Significant increase in the HDL-cholesterol level ($P<0.05$). Significant reduction in both the systolic and diastolic blood pressure ($P<0.001$). Non-significant effect on the weight and BMI (body mass index). Combined therapy with losartan and glibenclamide have produced significant effects on the carbohydrate metabolism, fat (only HDL-cholesterol level) metabolism and blood pressure of the hypertensive NIDDM patients.

INTRODUCTION

Diabetes is a chronic illness that requires continuing medical care and patient's self management education (Anonymous, 2003). Type-2 diabetes have tissue resistance to the action of insulin combined with a relative deficiency in insulin secretion (Martha *et al.*, 2004). It is a group of disorders characterized by hyperglycemia and associated with microvascular (optic, renal and possibly neuropathic) and macrovascular (coronary and peripheral vascular) complications (Isley *et al.*, 1999). Hypertension occurs in about 40% of patients with non-insulin dependent diabetes mellitus (Type-2 diabetes) that is to say diabetes mellitus and cardiovascular disease

originate from a common soil. Blood pressure and diabetes are the two independent risk factors for the premature cardiovascular disease. When they occur together the incidence of atherosclerosis, stroke and coronary artery disease increase by many folds. It may be appropriate to say, "diabetes is a cardiovascular disease" (Grundy *et al.*, 1999).

Renin angiotensin system (RAS) plays a pivotal role in the hypertensive diabetic patients because renin released from the juxtaglomerular apparatus of the kidney acts on its substrate angiotensinogen and generate angiotensin I which is converted by the ACE (angiotensin converting enzyme) into angiotensin II (Vaughan, 1998) which act

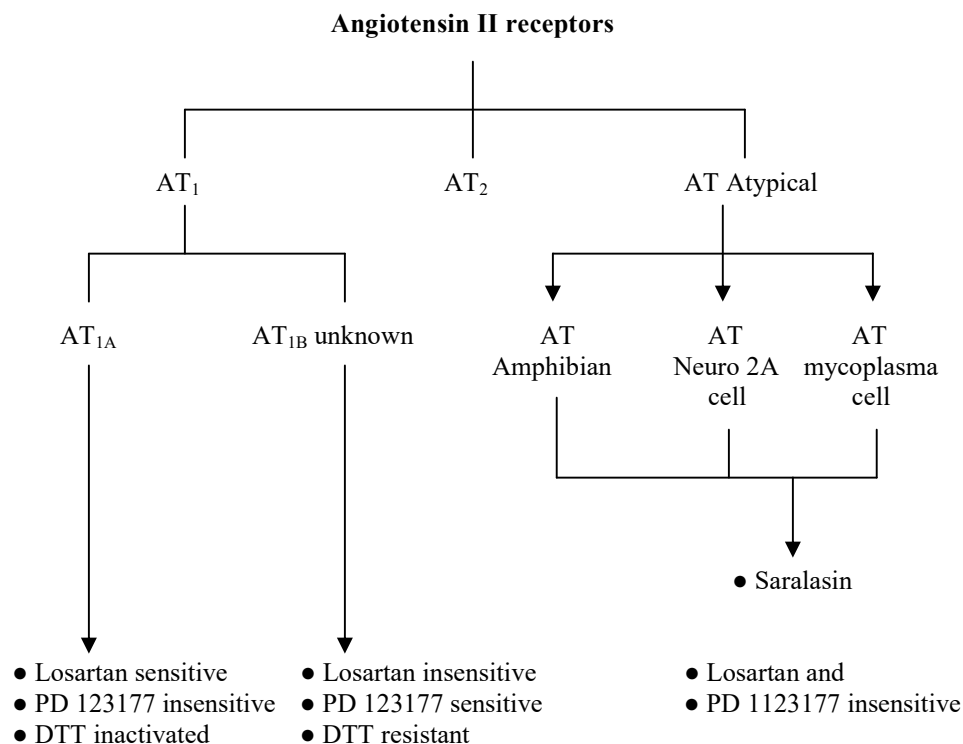


Fig. 1: Classification and characteristics of A-II receptors DTT, Dithiothreitol (Kang *et al.*, 1994).

through specific cell surface receptors. Angiotensin receptors can be differentiated into two distinct subtypes AT₁ and AT₂ besides AT atypical by radioligand-receptor binding studies using specific nonpeptide DuP 753 (losartan) and PD 123177 (Kang *et al.*, 1994) (Fig. 1). Angiotensin II so formed produces three major effects: 1) Altered peripheral resistance 2) Altered renal function and 3) Altered cardiovascular structure (Jackson, 2001).

Losartan is 2-N-Butyl-4-Chloro-5-hydroxymethyl-l-imidazole potassium salt. It is the most potent, orally active, specific, competitive angiotensin-II (A-II) receptor antagonist. It displaces A-II from its specific type I receptor subtype. This displacement antagonizes A-II induced smooth muscle contraction, aldosterone release, catechol-

amine release, arginine vasopressin release, water intake and hypertrophic responses (Bauer *et al.*, 1995) (Fig. 2).

Glibenclamide is a second generation oral hypoglycaemic agent belonging to the group of sulphonylureas. It acts on the B-cells of pancreas stimulating insulin secretion and thus reducing plasma glucose (Rang *et al.*, 2003). Their principle target is the ATP sensitive K⁺ channel (Gribble *et al.*, 1998). Sulfonylureas not only stimulates the insulin secretion from the B-cells but in addition they also increase the sensitivity of the peripheral tissues to insulin (Gaines *et al.*, 1988). Present study was designed to evaluate the effects of combined losartan and glibenclamide therapy on the carbohydrate (blood glucose and insulin level) and lipid metabolism in the hypertensive NIDDM patients.

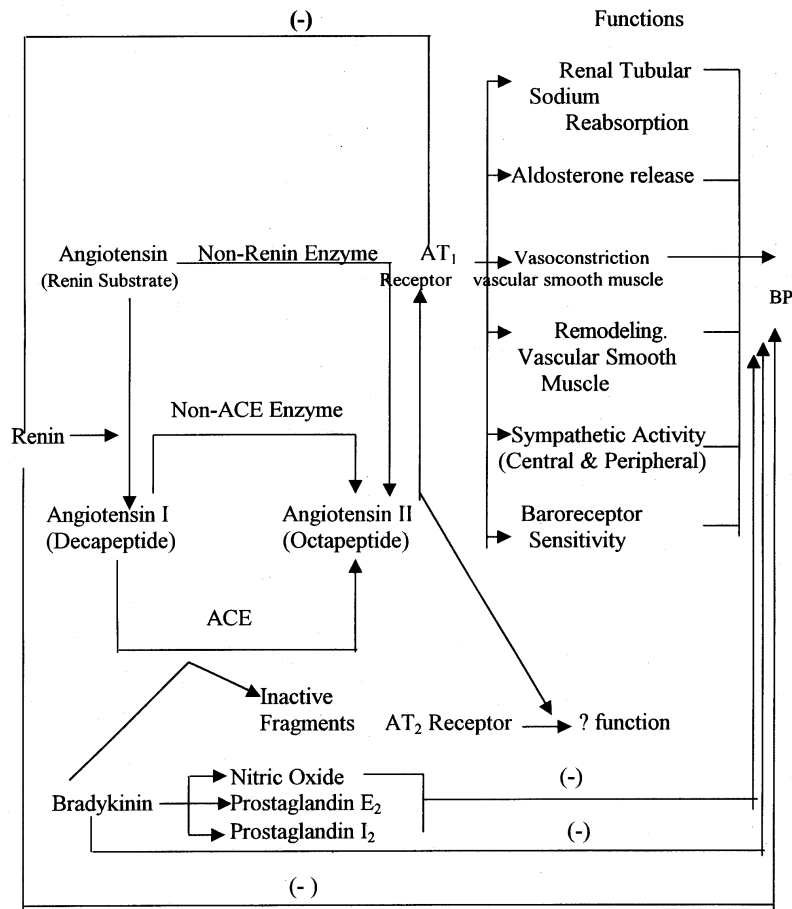


Fig. 2: Mechanism of action of ACE inhibitors and angiotensin II receptor blockers (Bauer *et al.*, 1995)

MATERIALS AND METHODS

A total of 10 newly diagnosed, untreated, uncomplicated hypertensive NIDDM patients were selected from the outpatient diabetic clinic of the Medical Ward III of JPMC, Karachi. Eligibility criteria included patients of both sexes with their ages ranging between 30-55 years, fasting blood sugar equal to or more than 140 mg/dl and blood pressure equal to or greater than 160/95 mmHg. Consent were obtained from all study participants before they were enrolled in the study. Preliminary data, date of follow up visit and laboratory investigations of each patient was recorded on

the specially designed proforma.

Study Design

The study was done for a period of 5 weeks. All patients were kept on the diet control (500 calories less than what was being taken) and exercise (session of walking 20 to 30 minutes daily) along with medication. Fasting blood sugar and blood pressure were taken once weekly whereas other investigations as serum insulin level, lipid profile (total cholesterol, triglycerides, HDL-cholesterol, LDL-cholesterol, VLDL cholesterol) were done twice during the study i.e. at day zero and day 35. A fixed dose of tablet losartan 50 mg once daily was given to

Table-1
Carbohydrate Metabolism (n=10)

Parameters	DAY-0	DAY-35	P-VALUE
Fasting blood glucose (mg/dl)	283.1 $\pm$ 22.93	156.80 $\pm$ 8.08	< 0.001
Fasting serum insulin (μ U/ml)	8.10 $\pm$ 1.51	17.32 $\pm$ 3.99	< 0.05

$\pm$ indicates standard error of mean.

Table-2
Fat Metabolism (n=10)

Parameters	DAY-0	DAY-35	P-VALUE
Total cholesterol	183.8 $\pm$ 11.71	209 $\pm$ 11.13	NS
Triglycerides	128.7 $\pm$ 17.12	173.7 $\pm$ 23.16	NS
HDL-cholesterol	35.3 $\pm$ 1.09	39.5 $\pm$ 1.43	< 0.05
LDL-cholesterol	114.1 $\pm$ 5.88	134.7 $\pm$ 9.11	NS
VLDL-cholesterol	25.74 $\pm$ 3.42	34.74 $\pm$ 4.63	NS

All observations measured in mg/dl.

$\pm$ indicates standard error of mean.

Table-3
Changes In Mean Systolic and Diastolic Blood Pressure Level,
Weight and Body Mass Index (n=10)

Parameters	DAY-0	DAY-35	P-VALUE
Blood pressure (mmHg)	* 171 $\pm$ 4.81	*134.5 $\pm$ 4.85	< 0.001
	**103 $\pm$ 2.13	**82 $\pm$ 2.0	< 0.001
Weight (Kg)	74.8 $\pm$ 4.28	72.1 $\pm$ 4.37	NS
Body mass index BMI (Kg/m ²)	30.84 $\pm$ 1.86	29.75 $\pm$ 1.90	NS

*systolic blood pressure, **diastolic blood pressure.

$\pm$ indicates standard error of mean.

each patient for 5 weeks. Tablet glibenclamide 5 mg (daonil) was given and adjusted according to the glycaemic control of the individual.

Analytical Methods

Fasting blood glucose concentration was measured with the one touch profile glucometer using the glucose oxidase method. Blood pressure was determined by sphygmomanometer. Serum insulin level was determined by the Microparticle enzyme immunoassay technique (MEIA) using the

IMX insulin reagent (NO.2A10-20) on the IMX system Abbott, USA (No. 8389-01). Lipid profile (total cholesterol, triglycerides, and HDL cholesterol) was done on the spectronic 21, spectrophotometer, Milton Roy Company, USA using the methods adopted by Allain in 1974, Bucoto in 1973 and Saunders Clinical diagnosis and management by laboratory methods 1984. LDL cholesterol was estimated by the Friedewald formula (1972).
 $CLDL = CPLASMA - CHDL - TG/5$
 VLDL cholesterol was calculated by formula proposed by Wilson cited by Delong *et al.*

(1990).

VLDL-cholesterol (mg/dl) = 0.20 X Triglycerides

RESULTS

In the carbohydrate metabolism marked decrease in the mean fasting blood glucose from 283.1 ± 22.93 mg/dl at day 0 to 156.8 ± 8.08 mg/dl at day 35 was observed. The decrease was statistically significant ($P < 0.001$) (Table-1). Mean serum insulin level also showed significant results ($P < 0.05$). It increased from 8.1 ± 1.51 μ U/ml at day 0 to 17.32 ± 3.99 μ U/ml at day 35 (Table 1).

In the fat metabolism non-significant ($P = NS$) increase was observed in the serum total cholesterol, serum triglyceride, serum LDL-cholesterol and serum VLDL-cholesterol levels which increased from 183.8 ± 11.71 mg/dl to 209 ± 11.13 mg/dl, 128.7 ± 17.12 mg/dl to 173.7 ± 23.16 mg/dl, 114.1 ± 5.88 mg/dl to 134.71 ± 9.11 mg/dl and 25.74 ± 3.42 mg/dl to 34.74 ± 4.63 mg/dl respectively from day 0 to day 35. In contrast to this serum HDL-cholesterol showed a significant increase ($P < 0.05$) from 35.3 ± 1.09 mg/dl at day 0 to 39.5 ± 1.43 mg/dl at day 35 (Table-2).

In the physiological parameters both the systolic and diastolic blood pressures showed significant reduction ($P < 0.001$) from 171 ± 4.81 mm of Hg to 134.5 ± 4.85 mm of Hg and 103 ± 2.13 mm of Hg to 82 ± 2.0 mm of Hg respectively from day 0 to day 35. As regards the weights of the patients and the body mass index a decline was observed in both of them but it was statistically insignificant (Table-3).

DISCUSSION

When Losartan was given in combination with glibenclamide to 10 hypertensive diabetic patients for 35 days mean fasting blood glucose level was decreased from 283.1 ± 22.93 mg/dl to 156.8 ± 8.08 mg/dl ($P < 0.001$) i.e. 44.619% decrease which is almost double when losartan was given alone ($P < 0.05$) and was 25.09% decrease (results of previous study conducted by the author herself).

Paolisso *et al* (1998) evaluated in 9 hypertensive patients after giving losartan 50 mg once daily for 4 weeks an enhanced glucose metabolic clearance rate from 5.1 ± 0.3 to 6.3 ± 0.4 mg/Kg x min ($P < 0.05$) and this is attributed to the sympathetic effect of losartan which increases the blood flow, inhibit glycolysis and thereby glucose clearance in the hypertensive patients. This study coincides with our results as there is reduction in blood glucose level. This reduction is highly significant probably due to the addition of the glibenclamides. Tomiyama *et al* (1994), conducted study in 18 spontaneously hypertensive rats with losartan for 3 weeks and found that glucose requirement for the clamp test did not changed i.e. no effect occurred on the blood glucose level. This contrast with our observations probably because our patients were diabetics and the animal model used was only hypertensive.

In our observations serum insulin level was increased from 8.1 ± 1.51 μ U/ml to 17.3 ± 3.99 μ U/ml at day 35 ($P < 0.05$ i.e. 113.8% $\uparrow$), when losartan glibenclamide combination was given. Study of Laakso *et al* (1996). Evaluated insulin sensitivity by the euglycemic clamp technique in 20 hyperinsulinemic subjects with essential hypertension when kept on losartan 50 mg for 12 weeks. They found no significant change in the serum insulin level. Our study does not favour these results may be because their patients were already hyperinsulinemic. Besides glibenclamide was used concurrently along with the losartan in our study and glibenclamides not only reduces the blood glucose level but also increases the insulin release.

In the lipid profile we have found significant results in the HDL-cholesterol level only which contrast with the study of Chan *et al* (1997) probably because his patients were elderly diabetics with albuminuria.

In our study both the systolic and diastolic blood pressures showed significant decrease from 171 ± 4.81 mm of Hg to 134.5 ± 4.85

mm of Hg and 103 ± 2.13 to 82 ± 2.0 mm of Hg respectively with $P < 0.001$. However, weight and body mass index showed non-significant results. MacKay *et al* (1996) and Moan *et al* (1996) have also found significant reductions in both the systolic and diastolic blood pressure. Chan *et al* (1997) have indicated non significant reduction in body mass index (Kg/m^2) after 12 weeks of therapy with losartan in 7 hypertensive NIDDM patients while Laakso (1996) have also observed non significant reduction in the body mass index (Kg/m^2) but in patients having essential hypertension. Losartan-glibenclamide therapy has exerted beneficial effects on the blood glucose, serum insulin, HDL-cholesterol and blood pressure of hypertensive NIDDM patients.

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